Customer Shopping Behavior Analysis

Uncovering insights from **3,900 purchases** to guide strategic business decisions

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Project Overview

This analysis examines real-world purchase data to reveal actionable patterns across three core dimensions of customer behavior.

Spending Patterns

Revenue by gender, shipping preference, and purchase frequency

Customer Segments

Loyal, returning, and new buyer profiles with subscription behavior

Product Preferences

Top-rated items, category trends, and discount sensitivity

Dataset at a Glance

3,900

Purchases

Total transactions analyzed

18

Data Columns

Rich feature set per record

37

Missing Values

Review Rating only — minimal impact

Demographics

Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Color

Shopping Behavior

Discounts, Promo Codes, Reviews, Shipping

Data Preparation Pipeline

A rigorous six-step process transformed raw data into a clean, analysis-ready dataset for SQL-powered insights.

01

Data Loading & Exploration

Initial inspection of structure, types, and distributions

02

Missing Data Handling

Median imputation applied per category for Review Rating

03

Column Standardization

Renamed all columns to snake_case for consistency

04

Feature Engineering

Created age_group and purchase_frequency_days derived fields

05

Data Consistency

Removed duplicate discount columns to eliminate redundancy

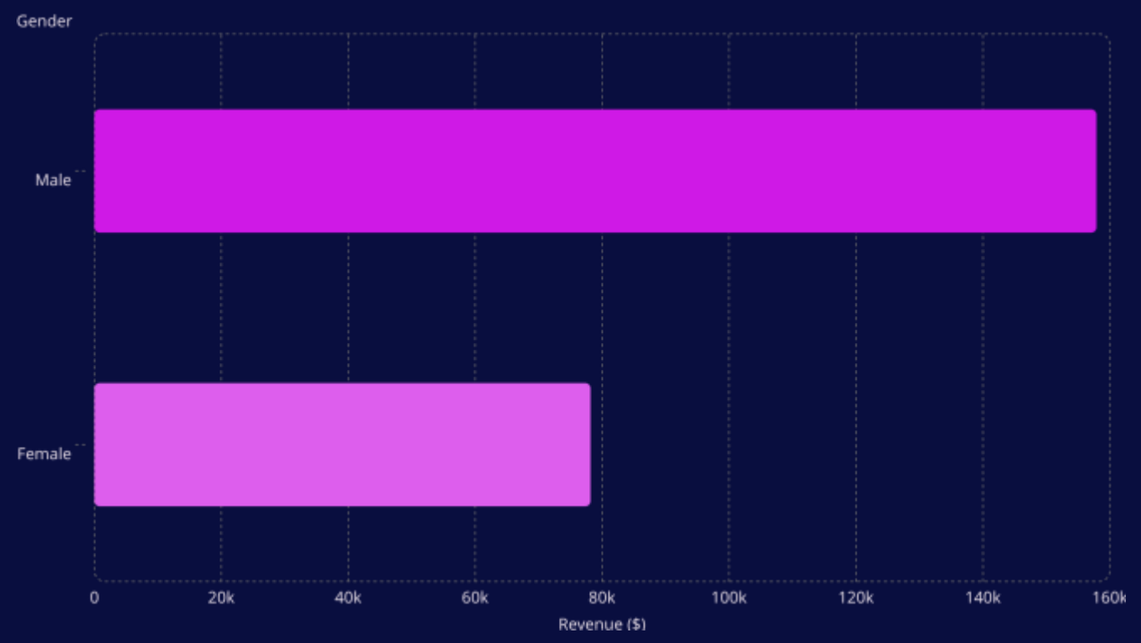
06

MySQL Integration

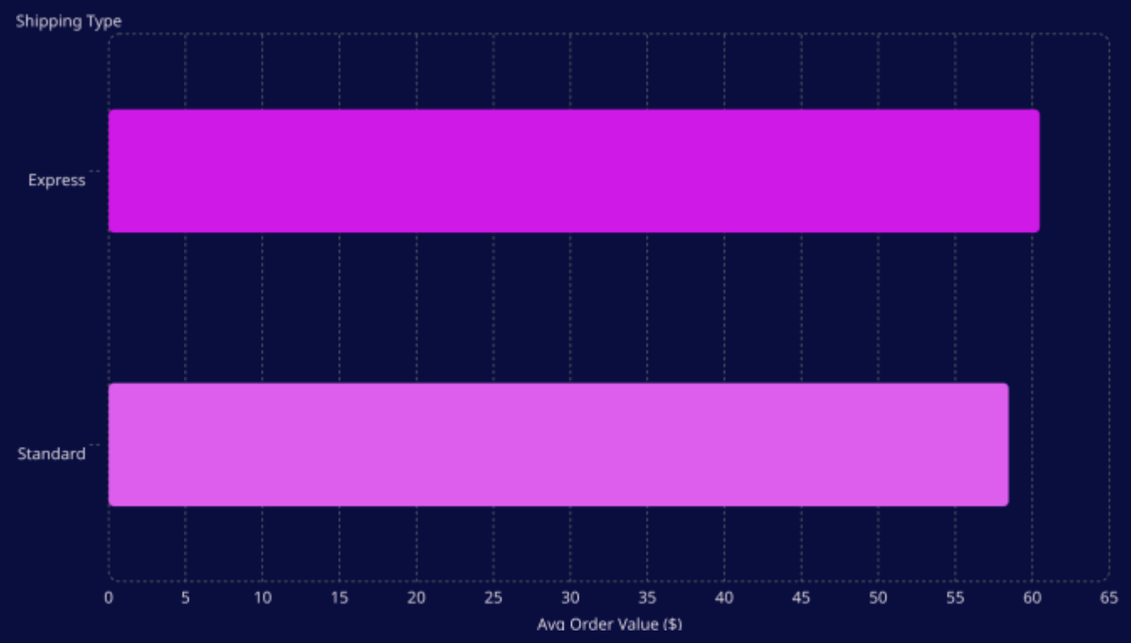
Loaded clean dataset into MySQL for structured SQL analysis

Revenue Insights

Revenue by Gender



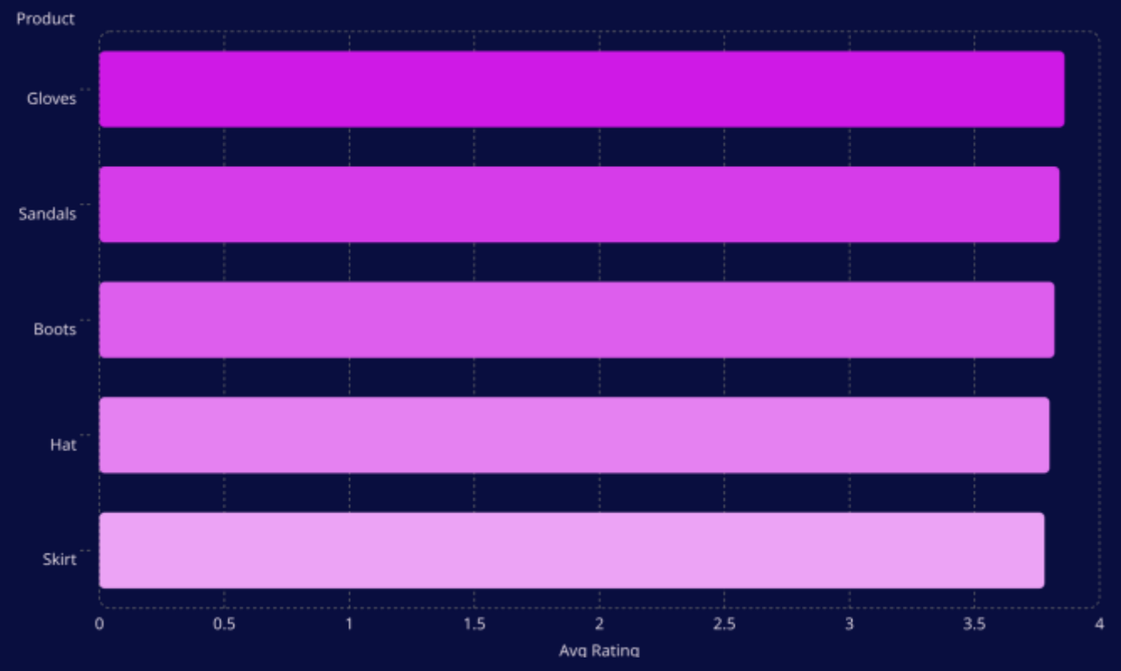
Average Order Value by Shipping Type



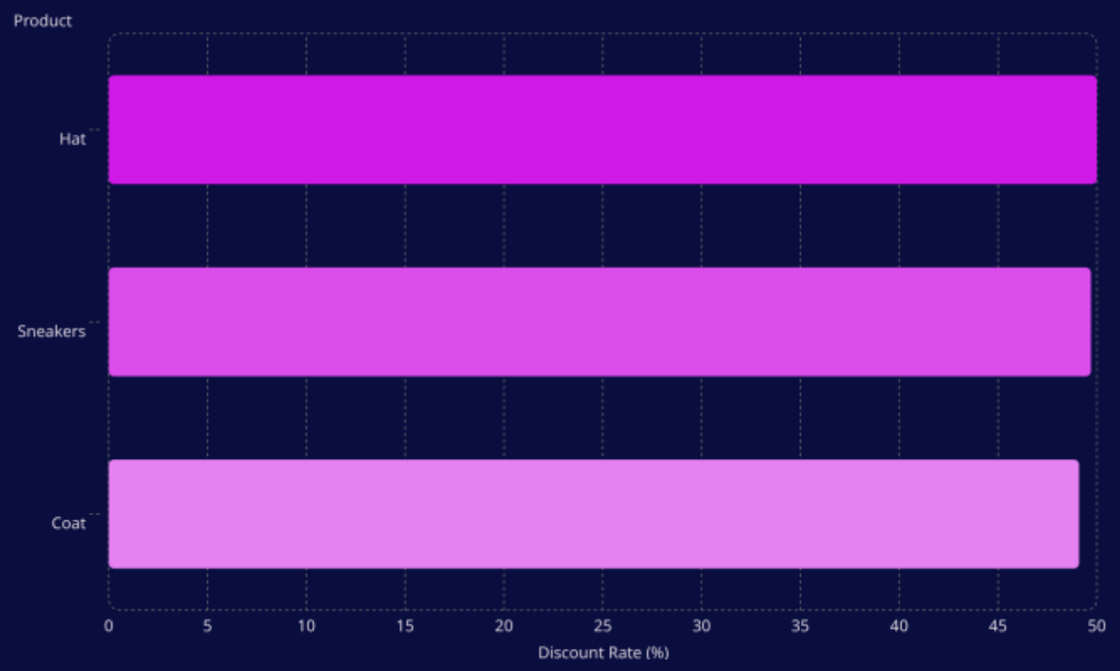
Key Insight: Male customers generate over **2x the revenue** of female customers. Higher-value customers also show a clear preference for **Express shipping** — a strong signal for premium targeting.

Product Performance

Top 5 Products by Review Rating



Discount-Dependent Products (% of sales with discount)



Hats, Sneakers, and Coats are **highly discount-sensitive** — nearly half of all sales require a promotion. Review ratings are tightly clustered, indicating consistent product quality across top performers.

Customer Segmentation

Segment Breakdown

The customer base is heavily weighted toward **loyal repeat buyers**, representing 79% of all purchasers. New customer acquisition remains a growth opportunity.

- **Loyal:** 3,116 customers — core revenue drivers
- **Returning:** 701 customers — strong re-engagement potential
- **New:** 83 customers — early-stage relationship

Subscribers

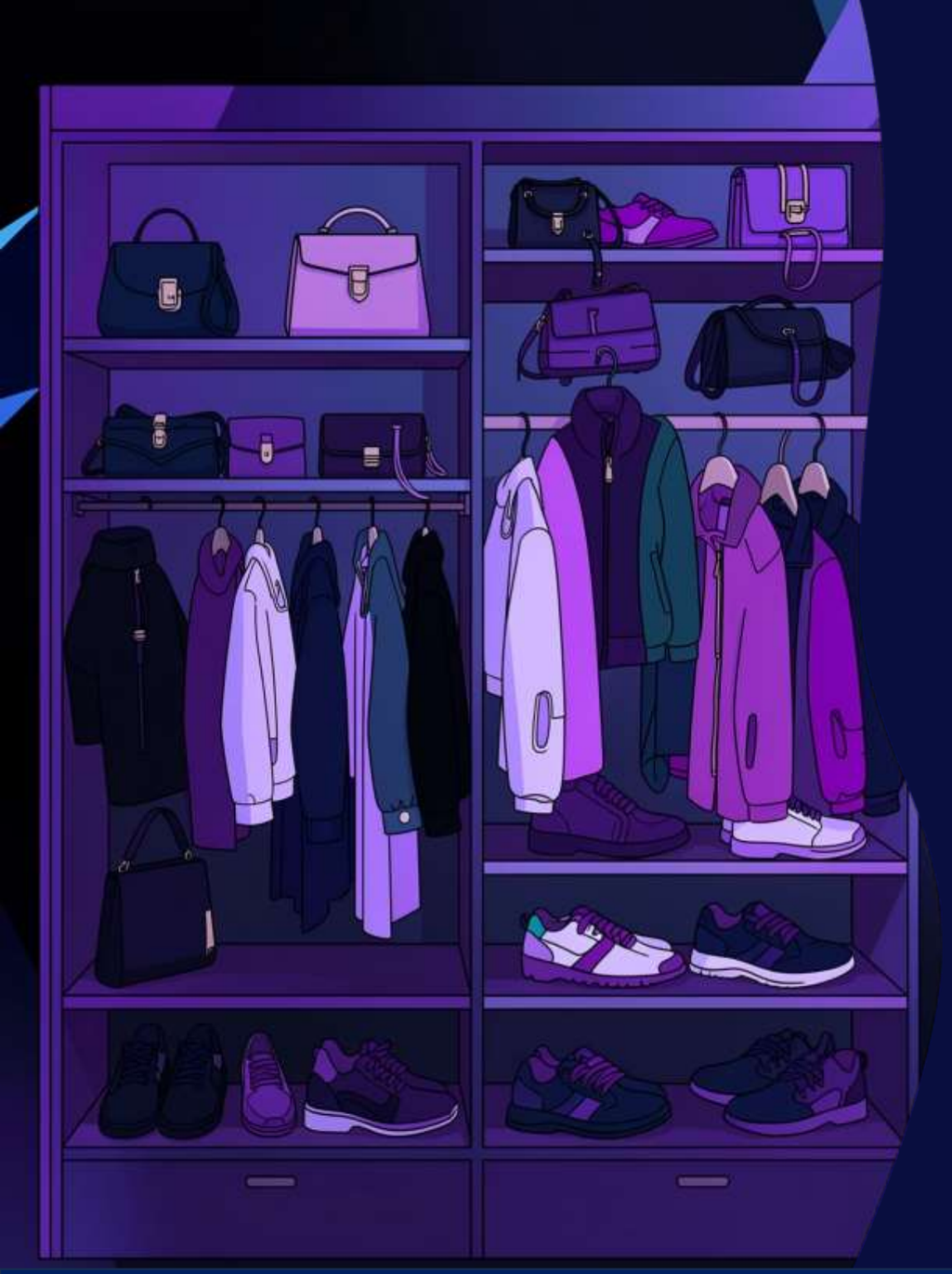
1,053 subscribers · Avg spend **$59.49**

Non-Subscribers

2,847 non-subscribers · Avg spend **$59.87**

Repeat Buyers

958 subscribers + **2,518** non-subscribers with 5+ purchases



Top Products by Category

Accessories

- Jewelry — 171 units
- Sunglasses — 161 units
- Belt — 161 units

Clothing

- Blouse — 171 units
- Pants — 171 units
- Shirt — 169 units

Footwear

- Sandals — 160 units
- Shoes — 150 units
- Sneakers — 145 units

Outerwear

- Jacket — 163 units
- Coat — 161 units

Key Business Recommendations

Boost Subscriptions

Offer exclusive perks to convert the 2,847 non-subscribers — a significant untapped revenue segment.

Loyalty Programs

Reward the 3,116 loyal customers to increase lifetime value and reduce churn risk.

Review Discount Policy

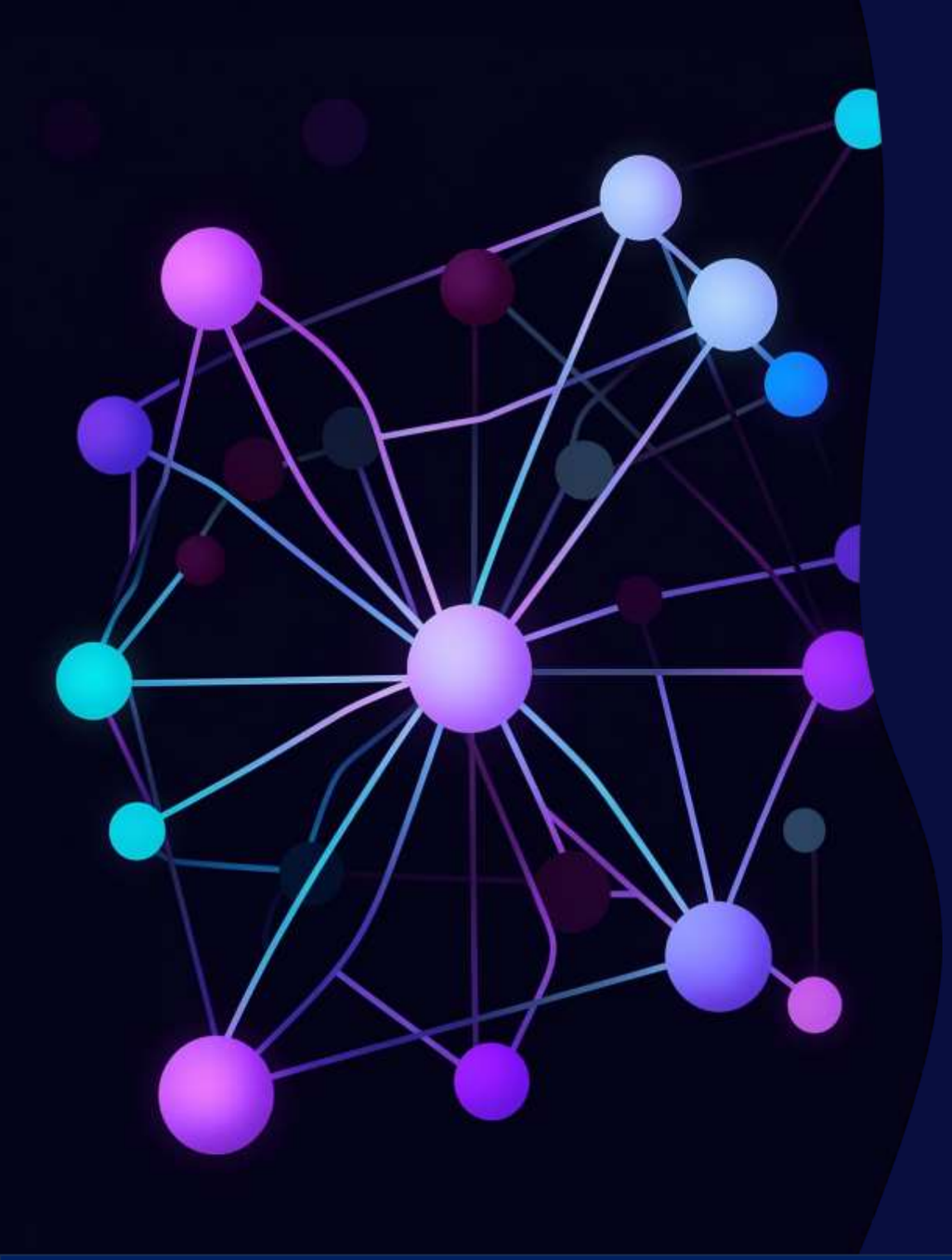
Control margins on Hat, Sneakers, and Coat — nearly 50% of sales rely on promotions.

Product Positioning

Feature top-rated products (Gloves, Sandals, Boots) prominently in campaigns to drive conversion.

Targeted Marketing

Focus spend on high-revenue age groups and Express shipping users for maximum ROI.



Thank You

This analysis of **3,900 purchases** reveals clear opportunities to grow revenue through smarter segmentation, subscription conversion, and targeted product strategy.

3,900 Purchases Analyzed	**5 Strategic Recommendations**
Ready to Act	

For questions or follow-up analysis, contact me.